

A Machine-Checked Decidability Result for the $\forall\text{PO}$ -Free Fragment of $\mathcal{ALCI}_{\text{RCC5}}$

Ordered-Disjoint Models, Finite Cone Schemes,
and Fresh-Occurrence Unfoldings

Michael Wessel
LambdaMikel@Home
miacwess@gmail.com

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Abstract

We study concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$, the description logic $\mathcal{ALCI}$ whose roles are the five jointly exhaustive, pairwise disjoint RCC5 relations, under strong equality and the RCC5 converse and composition laws. The full decidability problem remains open. We prove a sharp unconditional result for the fragment in which negation normal form contains no universal restriction $\forall\text{PO}.D$. Existential overlap, universal and existential disjointness, and both directions of proper-part modalities remain unrestricted; the fragment does not have the finite-model property.

The decision procedure is a finite greatest-fixed-point elimination on *cone signatures*. A signature records a closure type and a set of types that may occur strictly below it in the proper-part order. Completeness is obtained without a fragment assumption: the signatures realized by any RCC5 model form a post-fixed point of the elimination. For soundness, each surviving existential demand is unfolded to a fresh occurrence. Proper-part births generate a strict order, disjointness births generate a hereditary conflict relation, and every remaining pair is interpreted as partial overlap. This is an ordered-disjoint RCC5 frame by construction. The absence of $\forall\text{PO}$ is used exactly where the truth lemma would otherwise have to control all residual overlap edges.

The complete construction, including normalization of raw formulas, the polarity-sensitive fragment test, abstract-to-set representation, and an executable **Decidable** instance, is formalized in Lean 4 with no **sorry**. The formal capstones decide both abstract satisfiability and satisfiability by arbitrary nonempty sets. The explicit signature space is doubly exponential, so the result is presently a decidability theorem rather than a practical reasoner.

Keywords. Description logics; RCC5; qualitative spatial reasoning; modal logic; decidability; ordered-disjoint structures; greatest fixed points; Lean 4.

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1 Introduction

The Region Connection Calculus and the closely related Egenhofer relations provide a small qualitative vocabulary for spatial configurations. RCC5 classifies two nonempty regions as equal, a proper part in one direction or the other, partially overlapping, or disjoint. Since the five relations are jointly exhaustive and pairwise disjoint, they form a natural collection of modal roles. Cohn proposed modal spatial logics of this kind [3]; Wessel subsequently investigated the $\mathcal{ALCI}_{\text{RCC}}$ family and isolated the RCC5 and RCC8 cases as unresolved [9]. Lutz and Wolter later studied the corresponding modal logics systematically and again left the general/substructure RCC5 case open [6].

The difficulty is not merely that PP is transitive. Every two domain elements must receive exactly one RCC5 atom, and every ordered triple must satisfy the composition table. Thus a witness created for one modal demand immediately participates in constraints with all other witnesses. Moreover, concepts can force infinite proper-part chains, so a finite-model search is not complete. Familiar local blocking arguments are hazardous: identifying two equally

labelled occurrences can create an order cycle or move a disjointness witness into a cone where downward heredity makes the network inconsistent.

This paper isolates one polarity boundary at which those global problems can be controlled. In negation normal form we forbid only

$$\forall \text{PO}.D.$$

All existential modalities remain, including $\exists \text{PO}.D$, and all universal modalities over EQ, PP, PPI, and DR remain. On raw formulas the equivalent condition is polarity-sensitive: there is no positive $\forall \text{PO}$ and no negative $\exists \text{PO}$. This is necessary because $\neg \exists \text{PO}.D$ normalizes to $\forall \text{PO}.\neg D$.

1.1 Contributions

The main contributions are as follows.

1. We give an exact semantic account of strong atomic RCC5 frames as *ordered-disjoint structures*: a strict partial order together with a symmetric, irreflexive, downward-hereditary disjointness relation. Partial overlap is the residual case. We also give a canonical representation of every such structure by nonempty sets.
2. We define a finite cone-signature space and a monotone, reductive elimination operator. Iterating the operator for the size of the initial space reaches its greatest fixed point. A concept is accepted when a surviving signature contains it.
3. We prove completeness for *all* concepts. Given a model, each point contributes its closure type and the set of closure types realized by its strict proper parts. The model's realized signatures survive their own elimination and therefore survive every round.
4. We prove soundness for the $\forall \text{PO}$ -free fragment by a fresh-occurrence unfolding. No two existential demands share a generated element. Order and disjointness then satisfy the RCC5 frame conditions structurally, while the signature conditions propagate every permitted universal restriction.
5. We state the result both for NNF concepts and for formulas with unrestricted Boolean negation. We also prove equivalence between the abstract composition-table semantics and a direct semantics over arbitrary nonempty sets.
6. We identify the precise remaining obstruction to using the same construction for the full logic: a universal PO restriction observes not only explicit PO witnesses but every residual pair created by the ordered-disjoint completion.

All five steps are represented by named Lean declarations in the accompanying artifact. The purpose of this paper is to expose the mathematical spine of that development, not to reproduce a 45,000-line proof script.

1.2 Scope and non-claims

The result concerns satisfiability of one concept. It does not cover general TBoxes, ABoxes, nominals, number restrictions, role inclusions, role composition expressions in the object language, or the full logic with $\forall \text{PO}$. Because the fragment is not closed under negation, decidability of its satisfiability problem does not automatically yield a same-fragment subsumption procedure via $C \sqcap \neg D$. Nor do we claim a tight complexity classification or an efficient implementation.

2 Syntax and RCC5 semantics

2.1 The RCC5 relation algebra

Let

$$\text{Rel} = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}, \text{DR}\}.$$

The converse operation fixes EQ, PO, DR and exchanges PP with PPI. Write $\mathcal{U} = \text{Rel}$. The composition table used throughout is shown in Table 1; an entry gives the possible relation between x and z when the row relation holds from x to y and the column relation from y to z .

comp	EQ	PP	PPI	PO	DR
EQ	{EQ}	{PP}	{PPI}	{PO}	{DR}
PP	{PP}	{PP}	$\mathcal{U}$	{PP, PO, DR}	{DR}
PPI	{PPI}	{EQ, PP, PPI, PO}	{PPI}	{PPI, PO}	{PPI, PO, DR}
PO	{PO}	{PP, PO}	{PPI, PO, DR}	$\mathcal{U}$	{PPI, PO, DR}
DR	{DR}	{PP, PO, DR}	{DR}	{PP, PO, DR}	$\mathcal{U}$

Table 1: The atomic RCC5 composition table.

An *abstract RCC5 frame* is a nonempty set W with a total map $\rho : W^2 \rightarrow \text{Rel}$ such that, for all $x, y, z \in W$:

$$\begin{aligned} \rho(x, x) &= \text{EQ}, \\ \rho(x, y) = \text{EQ} &\implies x = y, \\ \rho(y, x) &= \rho(x, y)^\smile, \\ \rho(x, z) &\in \text{comp}(\rho(x, y), \rho(y, z)). \end{aligned}$$

Totality of ρ encodes joint exhaustiveness and functionality encodes pairwise disjointness of the base relations. Equality is therefore *strong*: EQ is the identity relation, not an equivalence later to be quotiented.

2.2 Concepts and satisfaction

We use the standard negation-normal-form grammar [1]:

$$C, D ::= \top \mid \perp \mid A \mid \neg A \mid C \sqcap D \mid C \sqcup D \mid \exists r.C \mid \forall r.C, \quad r \in \text{Rel}.$$

An interpretation adds a valuation $A^{\mathcal{I}} \subseteq W$ for every concept name. Boolean constructors have their usual meaning and

$$\begin{aligned} x \models \exists r.C &\iff \text{some } y \in W \text{ has } \rho(x, y) = r \text{ and } y \models C, \\ x \models \forall r.C &\iff \text{every } y \in W \text{ with } \rho(x, y) = r \text{ satisfies } C. \end{aligned}$$

Concept satisfiability asks whether $x \models C$ for some point of some abstract RCC5 interpretation.

The formalization permits a carrier type with an explicit domain predicate; all frame laws and modal quantifiers are relativized to that predicate. This is definitionally equivalent to passing to the subtype of in-domain elements, and nonemptiness is part of satisfiability.

2.3 The exact fragment

Definition 2.1 ($\forall\text{PO}$ -free NNF). The predicate $\text{POFree}(C)$ is defined structurally. It recurses through all Boolean and modal subformulas, permits every existential restriction, and permits $\forall r.D$ exactly when $r \neq \text{PO}$ and $\text{POFree}(D)$.

Thus “PO-free” would be a misleading abbreviation: overlap itself is not removed. The fragment freely contains $\exists\text{PO}.D$ and may mix such demands with arbitrary nesting of PP, PPI, and DR modalities.

For raw formulas with general negation, define a polarity $b \in \{+, -\}$ while traversing the syntax. Negation flips b ; conjunction and disjunction keep it. A positive universal PO modality and a negative existential PO modality are forbidden. Equivalently,

$$\text{FPOFree}^+(F) \implies \text{POFree}(\text{nnf}(F)).$$

This is the implication used by the decision wrapper. Notice that $\neg\forall\text{PO}.A$ is admitted, since its NNF is $\exists\text{PO}.\neg A$, whereas $\neg\exists\text{PO}.A$ is excluded.

Example 2.2 (Composition-forced inconsistency). Let

$$C_{\text{force}} = \exists\text{PP} . (\exists\text{DR}.A) \sqcap \forall\text{DR} . \neg A.$$

If $x\text{PP}y$, $y\text{DR}z$, and $z \models A$, then $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ forces $x\text{DR}z$, contradicting the universal at x . Hence C_{force} is unsatisfiable even though it belongs to the fragment. A correct procedure must therefore reason with more than the locally requested edges.

Example 2.3 (Failure of the finite-model property). The fragment concept

$$C_{\infty} = \exists\text{PP} . \top \sqcap \forall\text{PP} . \exists\text{PP} . \top$$

is satisfiable on the natural-number chain, with $m\text{PP}n$ iff $m < n$. At every PP-successor the universal restriction creates another strict successor. Transitivity makes these points pairwise distinct, so no finite model can satisfy C_{∞} .

3 Ordered-disjoint normal form

The key simplification is that the composition table has a compact structural presentation.

Definition 3.1 (Ordered-disjoint structure). An ordered-disjoint structure is a triple $(N, <, \#)$ such that:

1. $<$ is an irreflexive, transitive relation;
2. $\#$ is symmetric and irreflexive;
3. $x < y$ implies $\neg(x\#y)$; and
4. disjointness is downward hereditary in both coordinates:

$$x\#y, \quad x' \leq x, \quad y' \leq y \implies x'\#y',$$

where $\leq$ is the reflexive closure of $<$.

It induces an atomic network by priority:

$$\rho_{<,\#}(x, y) = \begin{cases} \text{EQ} & x = y, \\ \text{PP} & x < y, \\ \text{PPI} & y < x, \\ \text{DR} & x\#y, \\ \text{PO} & \text{otherwise.} \end{cases}$$

The incompatibility conditions make these cases exclusive. Partial overlap is not separately constructed; it is the residual relation left after identity, comparability, and disjointness have been ruled out.

Proposition 3.2 (Ordered-disjoint frame theorem). *Every ordered-disjoint structure induces an abstract RCC5 frame. Conversely, every abstract RCC5 frame yields an ordered-disjoint structure by defining $x < y$ iff $\rho(x, y) = \text{PP}$ and $x \# y$ iff $\rho(x, y) = \text{DR}$; the induced network then recovers ρ .*

Proof sketch. For the forward direction, identity and converse are immediate. The only composition cells that force a unique nonidentity output are consequences of the two structural closure conditions: $\text{PP}; \text{PP}$ and $\text{PPI}; \text{PPI}$ use transitivity, while $\text{PP}; \text{DR}$ and $\text{DR}; \text{PPI}$ use downward heredity. In all other cells the definition of the residual relation and the exclusion axioms place the output among the listed alternatives.

For the converse, $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ gives transitivity. Symmetry of DR follows from converse. Downward heredity is obtained by two singleton compositions, first $\text{PP}; \text{DR} = \{\text{DR}\}$ and then $\text{DR}; \text{PPI} = \{\text{DR}\}$. Strong equality and the fact that ρ is a function give irreflexivity and the separation of order from disjointness. A case split over the five possible values recovers the original network. $\square$

This normal form is more than convenient notation. It allows the soundness construction to prove a small family of geometric invariants rather than all 125 ordered-triple cases directly.

3.1 Canonical representation by sets

Ordered-disjoint structures also give a direct representation theorem. Let $\leq$ be the reflexive closure of $<$. For each $x \in N$, define a set of pairs

$$\eta(x) = \{(u, v) \in N^2 : \neg(u \# v) \text{ and } (u \leq x \text{ or } v \leq x)\}.$$

The pair (x, x) shows that $\eta(x)$ is nonempty. The defining axioms imply

$$\begin{aligned} \eta(x) \subseteq \eta(y) &\iff x \leq y, \\ \eta(x) \cap \eta(y) = \emptyset &\iff x \# y. \end{aligned}$$

Antisymmetry of $\leq$ makes η injective. Consequently equality, proper subset, proper superset, disjointness, and the remaining overlap case between the sets $\eta(x)$ agree exactly with $\text{EQ}, \text{PP}, \text{PPI}, \text{DR}$, and PO .

Theorem 3.3 (Abstract/set equivalence). *For every $\mathcal{ALCI}_{\text{RCC5}}$ concept C , with no fragment assumption,*

$$\begin{aligned} &C \text{ is satisfiable in an abstract RCC5 frame} \\ &\iff C \text{ is satisfiable by a pairwise-distinct family of nonempty sets} \end{aligned}$$

when RCC5 is interpreted by equality, proper inclusion in either direction, disjointness, and residual overlap.

The right-to-left direction reads an ordered-disjoint structure off an actual set family. The left-to-right direction reads an ordered-disjoint structure off the abstract frame and applies η . Both directions, including the satisfaction transport to the in-domain subtype, are formalized.

Remark 3.4 (Regular-closed regions). The Lean theorem concerns arbitrary sets; it does not formalize topology. Lutz and Wolter’s representation theorem states that every countable general RCC5 structure has an isomorphic representation by nonempty regular-closed sets in $\mathbb{R}^n$, for every fixed $n > 0$ [6, Theorem 23]. The unfolding constructed below is countable, because its occurrences are finite words over $\text{Rel} \times \mathbb{N}$. Combining the two results externally therefore transfers the fragment theorem to *substructure semantics* over arbitrary families of regular-closed regions in $\mathbb{R}^n$. This does not establish the corresponding claim for a full structure containing every regular-closed region: universal modalities are sensitive to adding regions.

3.2 Relationship to split-forest models

The ordered-disjoint normal form is conceptually close to the original split-forest idea, but the two notions live at different levels. Both expose the same semantic backbone: PP is an order, DR is hereditary down that order, and PO fills the pairs left over. Moreover, the soundness model used here really does have a split-forest flavour. Each non-EQ existential demand creates a fresh occurrence; PP and PPI births generate vertical order components, while DR and PO births begin new components. The result is an occurrence tree whose transitive vertical closure is read as an ordered-disjoint model.

The difference is substantial. An ordered-disjoint structure may have an arbitrary partial order and is an exact semantic normal form, whereas a split-forest model is a formula-relative presentation obtained by duplicating or splitting occurrences so that the relevant part of that order has a tree-like cover with bounded interfaces. The cone-scheme completeness proof does not extract such a forest from a given model: it records only a point's type and the spectrum of types below it. Tree shape enters later, solely in the soundness unfolding. Thus the present method validates the semantic core of the split-forest intuition while avoiding its hardest full-logic burden, namely a finite interface that controls every cross-branch relation. When $\forall\text{PO}$ is restored, splitting or unravelling creates new residual PO pairs that a universal overlap restriction can observe; ordered-disjointness alone does not show that their target types are coherent. This is why the normal form is a natural starting point for future split-forest work, but is not by itself a full-logic decidability proof.

4 The finite cone scheme

Fix an NNF concept C_0 . Let $\text{cl}(C_0)$ be its list of subformulas, including C_0 itself. The implementation intentionally permits repeated entries; all cardinality estimates below are upper bounds in terms of the list length $m = |\text{cl}(C_0)|$.

4.1 Types, cones, and static admissibility

A *candidate type* is any sublist T of $\text{cl}(C_0)$. Hence at most 2^m candidates are enumerated. These are deliberately partial Hintikka types, not maximal consistent sets. A Boolean support test requires:

- $\perp \notin T$ and no pair $A, \neg A$ occurs;
- $C \sqcap D \in T$ implies $C, D \in T$;
- $C \sqcup D \in T$ implies $C \in T$ or $D \in T$; and
- $\forall \text{EQ}.D \in T$ or $\exists \text{EQ}.D \in T$ implies $D \in T$.

The last clause uses strong equality: the only EQ-neighbour of a point is the point itself.

A *cone signature* is a pair

$$q = (T, S),$$

where T is the support type at a point and S is a set of candidate types allowed strictly below it in the PP order. Write

$$\text{cone}(q) = \{T\} \cup S.$$

Besides requiring T and all members of S to pass the support test, static admissibility requires, for every $U \in S$,

$$\{D : \forall \text{PP}.D \in U\} \subseteq T, \tag{1}$$

$$\{D : \forall \text{PPI}.D \in T\} \subseteq U. \tag{2}$$

These clauses are the type-level shadow of universal propagation upward and downward through the proper-part order.

The finite set $S_0 = \text{Sig}_0(C_0)$ consists of all statically admissible pairs drawn from

$$\text{Types}(C_0) \times \mathcal{P}(\text{Types}(C_0)).$$

Consequently,

$$|S_0| \leq 2^m 2^{2^m} = 2^{m+2^m}. \quad (3)$$

4.2 Transition compatibility

For a demand $\exists r.D \in T$, a target signature $q' = (T', S')$ must contain D in T' . The additional compatibility condition depends on r and is summarized in Table 2. Put $A_r(T) = \{E : \forall r.E \in T\}$.

Demand	Additional condition on $q = (T, S)$ and $q' = (T', S')$
$\exists \text{PP}.D$	$\text{cone}(q) \subseteq S'$
$\exists \text{PPI}.D$	$\text{cone}(q') \subseteq S$
$\exists \text{DR}.D$	for all $U \in \text{cone}(q)$ and $V \in \text{cone}(q')$, $A_{\text{DR}}(U) \subseteq V$ and $A_{\text{DR}}(V) \subseteq U$
$\exists \text{PO}.D$	$A_{\text{PO}}(T) \subseteq T'$ and $A_{\text{PO}}(T') \subseteq T$
$\exists \text{EQ}.D$	handled locally; no transition is generated

Table 2: The compatibility test. In every non-EQ row, $D \in T'$ is also required.

The PP clause says that everything at or below the source is strictly below the target. Together with Equations (1) and (2), this propagates vertical universals along arbitrarily long chains. The DR clause reflects downward heredity in both coordinates. The PO clause propagates direct universal PO obligations across an explicit overlap witness in both directions; although vacuous on the fragment, it is needed for the full-language completeness theorem discussed in Section 7.1.

4.3 Elimination and the decision test

Given a set X of signatures, define

$$\text{Prune}(X) = \{q \in X : \text{every non-EQ existential demand in } q \text{ has a compatible target in } X\}.$$

This operator is reductive and monotone:

$$\text{Prune}(X) \subseteq X, \quad X \subseteq Y \implies \text{Prune}(X) \subseteq \text{Prune}(Y).$$

Starting with S_0 , set $S_{i+1} = \text{Prune}(S_i)$. Since S_0 is finite and each strict change deletes at least one signature, $S_{|S_0|}$ is a fixed point. The decision test is

$$\text{Accept}(C_0) \iff \text{some } (T, S) \in S_{|S_0|} \text{ has } C_0 \in T. \quad (4)$$

Equivalently, the executable procedure is:

1. compute $\text{cl}(C_0)$ and all candidate types;
2. enumerate and filter the static cone signatures;
3. apply Prune exactly $|S_0|$ times; and
4. answer SAT iff a remaining support type contains C_0 .

All tests use finite lists and Boolean equality. The proof obligations are that no genuine model is eliminated and that every accepted fragment concept has a model.

5 Completeness: models form post-fixed points

The completeness proof is short because the finite state space is fixed before a model is considered. Let $\mathcal{I}$ be an arbitrary abstract RCC5 model. For $x \in W$, define its closure type

$$\text{tp}(x) = \{C \in \text{cl}(C_0) : x \models C\}$$

and its strict down-spectrum

$$\text{down}(x) = \{\text{tp}(y) : y\text{PP}x\}.$$

The model signature of x is

$$q_x = (\text{tp}(x), \text{down}(x)).$$

Lemma 5.1 (Static admissibility). *For every in-domain point x , $q_x \in S_0$.*

Proof sketch. Truth immediately gives the Boolean support clauses. If $y\text{PP}x$ and $y \models \forall\text{PP}.D$, the actual PP edge gives $x \models D$. Conversely, if $x \models \forall\text{PPI}.D$, the same edge read in the inverse direction gives $y \models D$. These are exactly the two static vertical clauses. $\square$

Let

$$Z_{\mathcal{I}} = \{q_x : x \in W\} \cap S_0.$$

This is a finite set even when the model is infinite, because it is a subset of the already finite S_0 .

Lemma 5.2 (Model transitions are compatible). *If $x \models \exists r.D$ and y is a witnessing r -neighbour satisfying D , then q_y is compatible with q_x for the demand (r, D) .*

Proof sketch. Certainly $D \in \text{tp}(y)$. If $x\text{PP}y$, then every point at or strictly below x is strictly below y by transitivity, giving the PP cone inclusion. The PPI case is dual. If $x\text{DR}y$, every pair chosen from the two down-cones is disjoint by downward heredity, so DR universals cross in both directions. If $x\text{PO}y$, symmetry of PO propagates direct $\forall\text{PO}$ bodies both ways. The EQ case is local because $x = y$. $\square$

Lemma 5.3 (Survival). $Z_{\mathcal{I}} \subseteq \text{Prune}(Z_{\mathcal{I}})$.

Proof. Take $q_x \in Z_{\mathcal{I}}$ and any non-EQ existential in its support type. The semantics supplies a witness y . By Lemma 5.2, $q_y \in Z_{\mathcal{I}}$ is a compatible target. Every demand therefore survives. $\square$

Monotonicity now does all remaining work. If $Z \subseteq S_0$ and $Z \subseteq \text{Prune}(Z)$, then induction gives $Z \subseteq S_i$ for every i . In particular every model signature survives the named final round.

Theorem 5.4 (Cone-scheme completeness). *For every NNF concept C_0 , with no POFree hypothesis, if C_0 is satisfiable then $\text{Accept}(C_0)$ holds. More strongly, at every round i some surviving signature contains C_0 .*

Proof. Choose a model point x satisfying C_0 . Then $C_0 \in \text{tp}(x)$, and Lemma 5.3 plus monotonicity keeps q_x in every S_i . $\square$

No bounded-width extraction, filtration quotient, or finite-model property is used here. The infinite model is consulted only to establish that one finite set of its realized signatures is a post-fixed point.

6 Soundness: the fresh-occurrence unfolding

Assume that $X \subseteq S_0$ is a fixed point in the one-sided sense $X \subseteq \text{Prune}(X)$, and choose $q_0 = (T_0, S'_0) \in X$ with $C_0 \in T_0$. For each non-EQ demand, fix the first compatible target in the finite list X . This deterministic choice turns X into a finite control graph.

6.1 Occurrences and births

An occurrence is a finite word over $\text{Rel} \times \mathbb{N}$. The empty word is the root. If the current signature's i th demand is (r, D) , prefixing (r, i) creates the child that serves it. Only words generated by this rule belong to the carrier. Different words are different domain elements; in particular, repeated visits to the same finite control state do not identify occurrences.

This freshness is what allows a finite control graph to describe an infinite model such as the one required by C_∞ . A cycle in the control graph unfolds to infinitely many distinct words rather than to a forbidden PP-self-loop.

6.2 The order

A PP birth points upward from a word w to $(\text{PP}, i) :: w$; a PPI birth points upward in the opposite direction, from $(\text{PPI}, i) :: w$ to w . Let $<$ be the transitive closure of these base steps. Define an integer height by adding $+1$ for a PP prefix, -1 for a PPI prefix, and 0 otherwise. Every base order step strictly raises height. Hence $<$ is irreflexive; its definition makes it transitive.

6.3 Disjointness

A DR birth is a seed pair between w and $(\text{DR}, i) :: w$. Define $\#$ as the symmetric downward closure of all such seeds with respect to $\leq$. It is symmetric and downward hereditary by definition.

To see why $<$ and $\#$ cannot collide, define the *vertical base* of a word by stripping all initial PP and PPI births. Order steps preserve the vertical base. A DR birth changes it, and every pair in $\#$ descends from the two different bases of such a birth. Consequently disjoint points have different vertical bases, while comparable points have the same base. This proves irreflexivity of $\#$ and $x < y \Rightarrow \neg(x \# y)$.

We have therefore constructed an ordered-disjoint structure on the generated occurrences. By Proposition 3.2, it induces a full RCC5 frame. Each birth has its advertised relation:

- PP and PPI births are base order edges;
- a DR birth is a disjointness seed; and
- a PO birth changes vertical base without being a DR seed, so its endpoints are unequal, incomparable, and nondisjoint; it is residual PO.

6.4 Labels and modal propagation

Label an occurrence with the support type of the control signature reached by following its word from q_0 . Fixed-point membership ensures that every generated step has a chosen target in X , and compatibility ensures that the target label contains the demanded body.

The universal cases follow the geometry of the unfolding:

- Along each proper-part step, compatibility pushes the entire source cone into the target's down-spectrum. This invariant composes along the transitive closure. Static admissibility then propagates $\forall\text{PP}$ upward and $\forall\text{PPI}$ downward.

- Every disjoint pair descends from a DR birth. Each endpoint label lies in the cone of the corresponding birth endpoint, and the DR compatibility clause propagates universal bodies across all such pairs.
- EQ is identity, so its universal and existential bodies are local support conditions.
- There is no propagation obligation for $\forall\text{PO}$, because no such subformula occurs in $\text{cl}(C_0)$ when $\text{POFree}(C_0)$ holds.

The last item is the exact point of the fragment restriction. The generated frame contains many PO edges that are not births: every unequal pair that is neither comparable nor disjoint becomes PO. Direct transition compatibility controls explicit PO witnesses, but it does not control all of these residual pairs. A $\forall\text{PO}$ formula would have to inspect every one of them.

Lemma 6.1 (Truth lemma). *Let C_0 be $\forall\text{PO}$ -free. Every concept in the label of a generated occurrence is true at that occurrence in the unfolded interpretation.*

Proof sketch. Induct on concepts. The support test discharges $\perp$, literal clashes, conjunction, disjunction, and the local EQ cases. A non-EQ existential uses its own fresh child and the exact birth relation. The PP, PPI, and DR universal cases use the propagation facts above. In the PO universal case, membership of $\forall\text{PO}.D$ in a label would imply its membership in $\text{cl}(C_0)$, contradicting $\text{POFree}(C_0)$; the case is therefore vacuous. $\square$

Theorem 6.2 (Cone-scheme soundness). *If C_0 is $\forall\text{PO}$ -free and a fixed point $X \subseteq S_0$ contains a signature whose support type contains C_0 , then C_0 is satisfiable in an abstract RCC5 frame. The witnessing frame is countable and is induced by an ordered-disjoint structure.*

Proof. Root the fresh-occurrence unfolding at the chosen signature and apply Lemma 6.1 to the empty word. $\square$

7 Decision theorems

Combining Theorems 5.4 and 6.2 at the named round gives the central characterization.

Theorem 7.1 (NNF concept satisfiability). *For every NNF concept C_0 with $\text{POFree}(C_0)$,*

$$C_0 \text{ is satisfiable} \iff \exists (T, S) \in S_{|S_0|} C_0 \in T.$$

Consequently, satisfiability of the $\forall\text{PO}$ -free fragment of $\mathcal{ALCI}_{\text{RCC5}}$ is decidable.

Because the right-hand side is a finite Boolean test, the formal result is a Lean **Decidable** value, not merely a classical excluded-middle theorem.

Theorem 7.2 (Raw formulas). *Let F be a formula with unrestricted Boolean negation. If F has no positive $\forall\text{PO}$ occurrence and no negative $\exists\text{PO}$ occurrence, then satisfiability of F is decidable.*

Proof sketch. The polarity-indexed normalization proves simultaneously that $\text{nnf}^+(F)$ is equivalent to F and $\text{nnf}^-(F)$ is equivalent to $\neg F$. The structural polarity condition implies $\text{POFree}(\text{nnf}^+(F))$. Apply Theorem 7.1 and transport the resulting decision across semantic equivalence. $\square$

Together with Theorem 3.3, the same algorithm decides satisfiability over pairwise-distinct families of arbitrary nonempty sets.

Corollary 7.3 (Concrete set satisfiability). *For every $\forall\text{PO}$ -free concept C_0 , satisfiability under the direct set interpretation of RCC5 is decidable.*

7.1 A one-sided theorem for the full language

Since completeness never used `POFree`, it yields a useful but limited corollary.

Corollary 7.4 (Sound full-language refutation). *For any NNF concept C_0 , if no signature surviving any chosen elimination round contains C_0 , then C_0 is unsatisfiable. In particular this test is sound at the named round $|S_0|$.*

The converse is not claimed. A surviving signature outside the fragment need not unfold to a model, because residual PO edges may violate a universal PO restriction. Thus Corollary 7.4 is a finite one-sided analyzer, not a decision procedure for full $\mathcal{ALCI}_{RCC5}$.

8 Formalization and verification boundary

8.1 Lean theorem map

The development uses Lean 4 core and represents all finite control structures as lists. Table 3 maps the mathematical results above to the capstone declarations in `formal/POFreeLift.lean` [10].

Mathematical role	Lean declaration
Ordered-disjoint structure induces an RCC5 frame	<code>odNet_frame</code>
Every abstract model reads off as ordered-disjoint	<code>odOfModel</code> , <code>odNet_odOfModel</code>
Static signatures and transition test	<code>sigStatic</code> , <code>compatB</code>
Reductive, monotone elimination and fixed point	<code>pruneSig_red</code> , <code>pruneSig_mono</code> , <code>coneScheme_gfp</code>
Model signatures form a post-fixed point	<code>modelSigs_survives</code> , <code>modelSigs_in_gfp</code>
Full-language completeness	<code>coneScheme_complete</code>
Generated ordered-disjoint carrier	<code>gOD</code> , <code>unfInterp_rcc5</code>
Universal propagation / existential service	<code>allPP_gLt</code> , <code>allPPI_gLt</code> , <code>allDR_gDisj</code> , <code>gchild_serves</code>
Fragment truth lemma and soundness	<code>unf_truth</code> , <code>coneScheme_sound</code>
Named-round characterization and NNF decision	<code>coneScheme_correct_at</code> , <code>decidableSat_cone</code>
Raw normalization and decision	<code>nnfP_correct</code> , <code>pofree_nnfP</code> , <code>decidableFSat</code>
Abstract/set equivalence and set decision	<code>satisfiable_iff_set</code> , <code>decidableSetSat</code>
One-sided full-language refutation	<code>coneScheme_unsat_full</code> , <code>coneScheme_unsat_full_at</code>

Table 3: Result-to-artifact map for the cone-scheme proof.

The source contains no `sorry`, user-declared axiom, unsafe definition, `native_decide`, or implementation override in the audited snapshot. The printed axiom footprint of the main capstones is `propext`, `Classical.choice`, and `Quot.sound`, the standard principles used by Lean’s classical reasoning and quotient infrastructure. Classical choice is used in model-side existence proofs and in a stabilization argument; the finite decision data itself is computed by Boolean list operations. In particular, `decidableSat_cone` and `decidableFSat` are executable definitions. The artifact includes small kernel-reduced positive and negative examples.

8.2 Audited snapshot and reproducibility

This manuscript describes repository commit `795b2707b72e927c56b465ef0413b77aeb9a66a6`. The relevant files in that snapshot have the following identifiers:

File	SHA-256
formal/POFreeLift.lean	5f85ff556d0d67205432311f136a903 35b4a0540576604ec0f395e4aa3db28ae
formal/RCC5NormalForm.lean	438e8277891054c82050d2d1c830651 00881b0630692b72c7c71a5ee80d127ad

The cumulative `POFreeLift.lean` file has 45,375 lines; much of it records earlier constructions, while the current cone scheme and its capstones occur near the end. Recorded independent audits built the development under Lean 4.32.0 and 4.33.1 with zero errors, warnings, and sorries. The minimal build command is:

```
lean POFreeLift.lean
```

The snapshot does not contain a `lean-toolchain` file or a Lake project. Before archival release, the repository should pin the tested Lean version and provide a minimal module containing only the theorem’s dependency closure. This is a reproducibility and reviewability improvement, not a logical premise of the theorem.

8.3 Adversarial testing

The formal proof is the correctness evidence; testing is supplementary. The repository nevertheless contains independent probes that rederive the RCC5 table from finite set models, compare the finite control layer with exhaustive small-model search, mutate compatibility conditions, and instantiate bounded prefixes of the unfolding. Three staged cold reviews focused respectively on the trusted semantic base, completeness, and soundness. The completeness review found that an early PO transition clause omitted symmetric $\forall$ PO propagation. The clause was strengthened, and its preservation of completeness and fragment soundness was then proved in Lean. The soundness-focused review traced apparent probe failures to hypotheses that the probe had abstracted away and led to corresponding probe corrections. No review found a counterexample to the fragment capstone.

These experiments should not be conflated with proof, and their finite-model or bounded-depth nature cannot validate the infinite unfolding. Their value is different: they exposed transcription errors, under-strength conditions, and misleading prose before release.

9 Complexity and practical status

Let $m = |\text{cl}(C_0)|$. Candidate types number at most 2^m , and Equation (3) gives at most 2^{m+2^m} raw signatures before static filtering and deduplication. A direct implementation checks, for every surviving signature and every existential demand, whether some target is compatible, and performs at most $|S_0|$ rounds. Polynomially many scans of an explicitly represented doubly-exponential state space remain within a coarse doubly-exponential time upper bound.

The Lean development proves finiteness and termination but does not formalize this asymptotic analysis, nor does it establish a matching lower bound. More importantly, the literal list implementation materializes a power set of the type space. Audit runs report that only very small closures are feasible and that a one-modality example can exhaust substantial memory. The procedure is therefore not a competitive reasoner.

There is considerable room for symbolic improvement. The second component of a signature is used through inclusions and universal-body summaries, suggesting antichain representations, binary decision diagrams, or SAT/SMT encodings. Such changes would require new correctness proofs: quotienting two spectra merely because they agree on currently inspected summaries risks reintroducing the same joint-coherence problem that freshness was designed to avoid.

10 Related work

RCC and qualitative spatial reasoning. RCC8 was introduced as a region-based spatial logic by Randell, Cui, and Cohn [7]; RCC5 is its coarser five-relation calculus. Renz and Nebel studied tractable relation-algebra fragments for RCC reasoning [8]. Those constraint-satisfaction results concern finite networks of spatial variables. The present modal satisfiability problem adds arbitrary nesting, unary concept predicates, and models that may need to be infinite, so finite RCC network consistency is an ingredient rather than a decision procedure.

Modal RCC5. Cohn proposed modal and non-modal qualitative spatial languages [3]. Wessel developed the $\mathcal{ALCI}_{\text{RCC5}}$ family and documented both decidable and undecidable neighbouring systems, leaving the RCC5 case open [9]. Lutz and Wolter formalized modal logics over RCC8 and RCC5 region structures [6]. Their general RCC5 structures have the same essential conditions used here: JEPD atoms, identity equality, converse, and the composition table. Their Theorem 23 supplies the regular-closed representation used in Remark 3.4. Their RCC5 undecidability theorem assumes a stronger least-superregion property and applies to full classes satisfying it; they explicitly leave other RCC5 logics, including the general/substructure case, open. It therefore does not subsume the fragment theorem presented here.

The syntactic cut considered here is also different from simply dropping the PO modality. Existential PO remains available; only universal access to all PO neighbours is removed after NNF. We are not aware of an earlier published result isolating this one-polarity fragment, but this narrow novelty claim should be checked independently in peer review.

Description logics with concrete domains. Concrete-domain extensions connect abstract individuals to concrete values by features and predicates. Tableau and admissibility results for such systems include Lutz and Milićić [5] and later work by Baader and Rydval [2]. They do not directly settle the present logic: here the region objects themselves are the modal domain and the RCC5 relations connect every pair of them.

Machine-checked metatheory. Lean 4 combines dependent type theory with an extensible elaborator and proof automation [4]. In this development it serves two roles: checking the finite fixed-point argument and checking the infinite model construction. The latter is the more consequential use, since the fresh-occurrence carrier, the order/disjointness invariants, and the modal truth lemma are all quantified over unbounded objects.

11 Why full $\mathcal{ALCI}_{\text{RCC5}}$ remains open

The current method draws a clean line between what is already solved and what is not. Completeness works for a full-language input because a real model supplies compatible witnesses. Soundness fails to generalize because a control transition sees only an explicit witness edge, whereas the induced ordered-disjoint frame assigns PO to every residual pair.

Consider a source occurrence labelled $\forall \text{PO}.D$. It is not enough to ensure D at the target of each explicit $\exists \text{PO}$ birth. The source may be residual-PO related to occurrences created several generations away, to siblings created for unrelated demands, or to points in other vertical components. Whether such a pair is residual depends jointly on the absence of identity, order in either direction, and hereditary disjointness. A finite control state must either predict the types of all those targets or impose a model shape in which their obligations can be summarized.

This observation suggests four concrete research directions.

1. **Symbolic residual neighbourhoods.** Enrich a signature with a finite summary of which types may be residual-PO related to its cone, and seek a monotone elimination whose post-fixed-point completeness can still be read directly from models.
2. **Formula-relative split forests.** Start from the ordered-disjoint normal form, split only the occurrences needed to obtain a tree-like cover, and carry bounded interfaces that explicitly propagate $\forall\text{PO}$ obligations across forgotten branches. The missing theorem is a completeness bound on those interfaces, not the existence of the underlying order/disjointness structure.
3. **Disjunctive composition information.** Cone signatures encode consequences of singleton-forcing cells particularly well. Full PO control may require finite records of multi-path or disjunctive constraints, together with a proof that the resulting operator remains monotone.
4. **A negative boundary result.** If no finite residual neighbourhood summary can be made complete, that obstruction may guide an undecidability reduction. Any such reduction must force the coincidence or coordination of independently generated witnesses; merely constructing a grid-shaped RCC5 model is insufficient.

The ordered-disjoint normal form should precede these attacks. It identifies exactly which parts of RCC5 are positive closure conditions and which part is the complement. The full problem is, in this sense, the logic of a modal operator over that complement.

12 Limitations and release recommendations

The mathematical result is unconditional within the stated semantics, but a responsible release should keep the following boundaries visible.

- The main theorem is concept satisfiability for the specified role and syntax, not ontology reasoning with general axioms.
- The direct Lean representation uses arbitrary nonempty sets. The regular-closed corollary relies on the external representation theorem and concerns substructures, not full topological region structures.
- The explicit algorithm is prohibitively expensive. It should not be advertised as an implemented practical reasoner, and no optimal complexity result is claimed.
- The repository is cumulative and contains superseded proof routes. The release should designate the cone-scheme capstones as normative, move obsolete theorem-level narratives out of the main path, and include the exact theorem-to-source map from Table 3.
- A pinned Lean toolchain, a minimal build target, archived build output, and a stable commit or DOI should accompany the preprint.
- Independent human review by researchers in description logic, qualitative spatial reasoning, and Lean remains desirable. Machine checking removes proof-script gaps relative to the definitions; it does not by itself establish that the definitions match every convention a reader may associate with the name $\mathcal{ACCT}_{\text{RCC5}}$.

Promising follow-up work within the decided fragment includes a symbolic implementation, an exact complexity classification, extension to restricted TBoxes, and a Lean formalization of the regular-closed representation bridge.

13 Conclusion

Removing universal partial-overlap restrictions after negation normalization leaves a substantial logic: existential overlap, full disjointness, both proper-part directions, arbitrary Boolean structure, and models with infinite chains all remain. Its satisfiability problem is nevertheless decidable.

The proof separates finiteness from model size. A finite cone scheme records only closure types and their possible strict predecessor types. Any genuine model yields a post-fixed point, which proves completeness without quotienting the model. Conversely, a surviving scheme is unfolded into fresh occurrences. Their vertical births and disjointness births form an ordered-disjoint structure, so RCC5 closure holds globally and existential demands never need to reuse a node. The only missing truth-lemma branch is $\forall PO$, exactly the constructor excluded by the fragment.

This establishes a machine-checked positive result while sharpening the open full-logic problem. Ordered-disjoint normal form is the semantic core; controlling the residual PO neighbourhoods is the remaining frontier.

Artifact availability and authorship transparency

The formalization, probes, review packets, and historical development are available at <https://github.com/lambdamikel/alcirc5>; this paper targets the commit identified in Section 8. The $\mathcal{ALCI}_{RCC5}$ research programme and split-forest idea originate with Michael Wessel. The 2026 project used multiple AI systems under his direction for proposing proof architectures, producing Lean code, testing, and adversarial review. In particular, GPT-5.6 Sol proposed the finite-control/fresh-occurrence cone-scheme architecture, and Claude systems carried out the principal Lean development. This manuscript was prepared with OpenAI Codex from the audited repository state. AI systems are acknowledged as tools rather than listed as authors; the human author remains responsible for checking the text, claims, and public release.

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A Formal capstone signatures

For reference, the central Lean interfaces in the audited snapshot are:

```
def POFree : Concept -> Prop

theorem coneScheme_complete
  (hI : RCC5Interp I) (C0 : Concept) {x : alpha}
  (hx : I.dom x) (hC0 : sat I x C0) (n : Nat) :
  exists q in gfpIter pruneSig (sigStatic C0) n,
    C0 in q.1

theorem coneScheme_sound
  (hXS : forall q in X, q in sigStatic C0)
  (hfix : forall q in X, q in pruneSig X)
  (hpo : POFree C0) (hq0 : q0 in X)
  (hC0 : C0 in q0.1) :
  Satisfiable C0

theorem coneScheme_correct_at (C0 : Concept)
  (hpo : POFree C0) :
  Satisfiable C0 <=>
    exists q in
      gfpIter pruneSig (sigStatic C0) (sigStatic C0).length,
        C0 in q.1

def decidableSat_cone (C0 : Concept)
  (hpo : POFree C0) : Decidable (Satisfiable C0)

def decidableFSat (F : Formula)
  (h : FPOFree true F) : Decidable (FSatisfiable F)

theorem satisfiable_iff_set (C0 : Concept) :
  Satisfiable C0 <=> SetSatisfiable C0

def decidableSetSat (C0 : Concept)
  (hpo : POFree C0) : Decidable (SetSatisfiable C0)
```